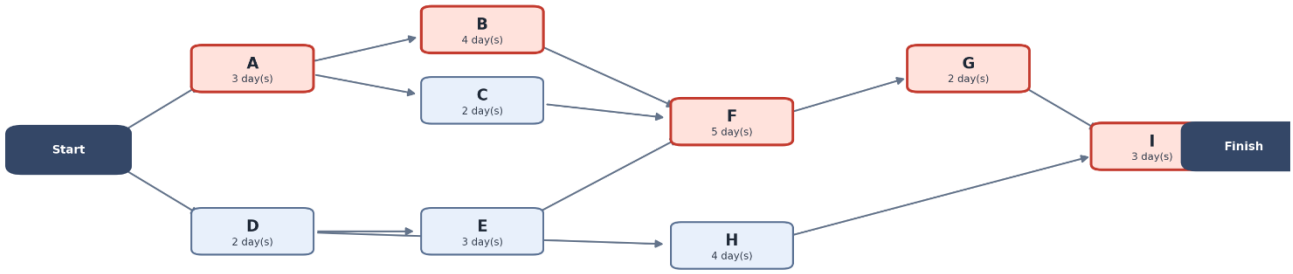
Task 1

GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

Activities, predecessors, and assumed durations

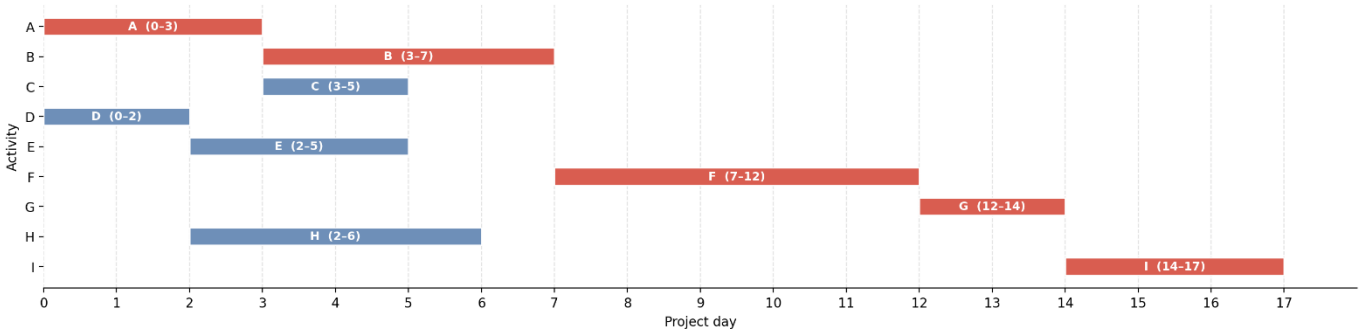
Activity	Predecessor	Duration (days)
A	—	3
B	A	4
C	A	2
D	—	2
E	D	3
F	B, C, E	5
G	F	2
H	D	4
I	G, H	3

PERT / Precedence Network (duration shown inside each activity)



Red outline = critical-path activities. This is an activity-on-node PERT network, so dummy activities are not required.

Gantt Chart | Estimated total project duration: 17 days



Red = critical path | Blue = non-critical

Task 2

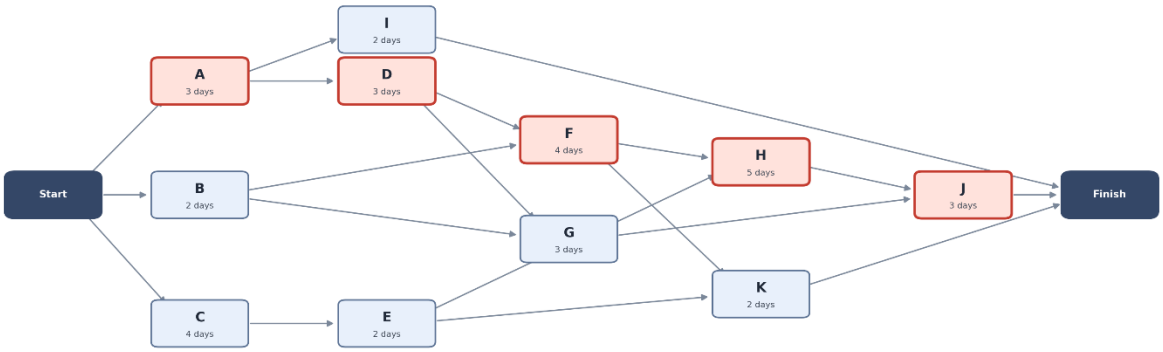
GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

PERT Network and Gantt Schedule

ACTIVITY TABLE

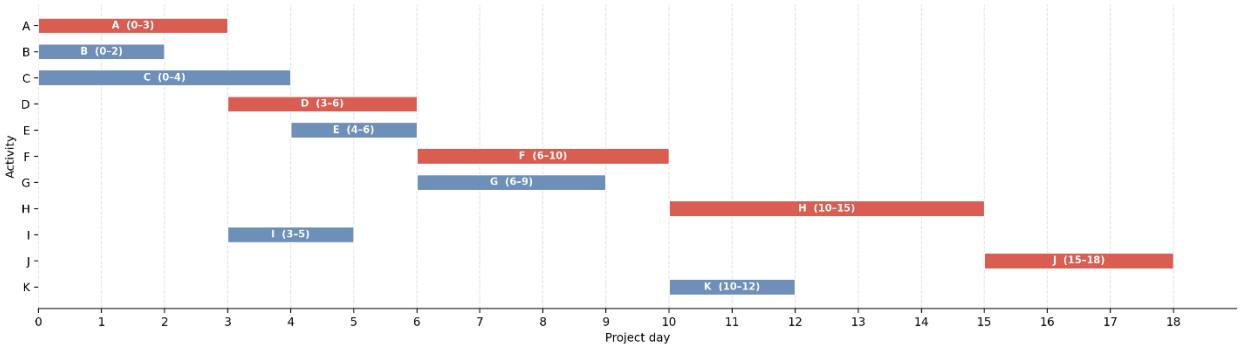
Activity	Predecessor	Duration (days)
A	—	**3**
B	—	**2**
C	—	**4**
D	A	**3**
E	C	**2**
F	B, D	**4**
G	B, D	**3**
H	E, F	**5**
I	A	**2**
J	G, H	**3**
K	E, F	**2**

PERT / PRECEDENCE NETWORK (duration inside each activity)



Red outline = critical-path activities. AON network: no dummy activities are needed.

GANTT CHART | ESTIMATED PROJECT DURATION: 18 DAYS



Red = critical path | Blue = non-critical activities

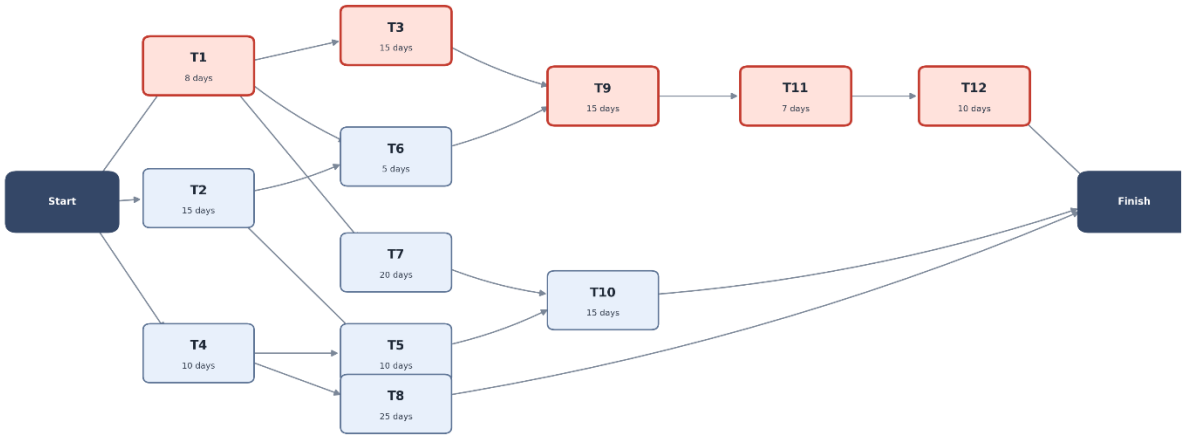
Task 3

GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

Activity Table — predecessor relationships and durations

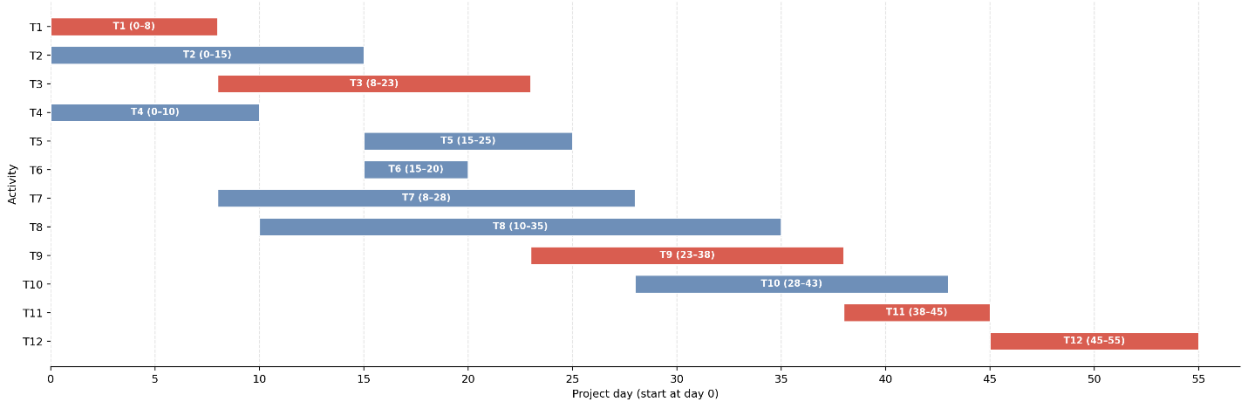
Activity	Duration (Days)	Predecessor
T1	8	—
T2	15	—
T3	15	T1
T4	10	—
T5	10	T2, T4
T6	5	T1, T2
T7	20	T1
T8	25	T4
T9	15	T3, T6
T10	15	T5, T7
T11	7	T9
T12	10	T11

PERT / PRECEDENCE NETWORK — duration inside each activity



Red outline = critical path. Activity-on-node (AON) format; dummy activities are not required.

GANTT CHART | ESTIMATED TOTAL DURATION: 55 DAYS



Red = critical-path activities | Blue = non-critical activities